Context & The Overfitting Problem

The Task: Character-level Sequence-to-Sequence (Seq2Seq) transliteration from Arabizi (Latin-script Arabic) to Arabic script.

The Challenge: Arabizi inherently contains "label noise" (e.g., a single Arabizi word can have multiple valid Arabic spellings).

The Baseline Problem: A standard 4-layer LSTM with Adam optimizer severely overfits to the training data, memorizing specific quirks rather than learning generalizable rules.

Baseline Metrics (10 Epochs):

Training: Exact Match (EM) = 65.31% | Character Error Rate (CER) = 7.96%

Testing: Exact Match (EM) = 51.17% | Character Error Rate (CER) = 15.04%

Generalization Gap: A massive ~14.1% drop in Exact Match accuracy from train to test.

How SAM Improves Generalization

Core Idea: Instead of just finding the lowest point in the loss landscape (which might be a sharp, narrow minimum that overfits), SAM finds a flat minimum that performs well even when weights are slightly perturbed.
The Mathematical Goal

Minimize the worst-case loss within a small weight neighborhood of radius ρ.

$\min_{w} \max_{\|\varepsilon\|_2 \le \rho} L(w + \varepsilon)$

The 2-Step Algorithm (per batch)

1. Perturb: Compute the gradient g at the current weights w. Move toward the sharpest nearby point.
$w' = w + \varepsilon$, where $\varepsilon = \rho g / \sqrt{(g_1^2 + g_2^2 + \cdots + g\square^2)}$

2. Restore & Step: Compute the gradient at w', restore the original weights w, and update using the base optimizer (Adam).

Trade-off: Requires two forward/backward passes per step (approximately 2× training time), but can improve robustness to label noise.

Performance Comparison & The Generalization Gap

Primary Success: SAM successfully acted as a regularizer, significantly shrinking the generalization gap.
Secondary Observation: Absolute test performance dipped slightly, indicating a shift from severe overfitting to slight underfitting.

Metric	Baseline (Adam)	SAM ($\rho=0.05$)	Change
Train Exact Match	65.31%	54.62%	▼ 10.69 pts *(Less memorization)*
Test Exact Match	51.17%	45.23%	▼ 5.94 pts
Train CER	7.96%	12.15%	▲ 4.19 pts
Test CER	15.04%	17.26%	▲ 2.22 pts
EM Generalization Gap	~14.14 pts	9.39 pts	▼ 4.75 pts (Improved)
CER Generalization Gap	~7.08 pts	5.11 pts	▼ 1.97 pts (Improved)

Diagnosing the SAM Trade-off

While the shrinking gap proves SAM is working as intended, the drop in absolute test accuracy is driven by three likely factors:

Aggressive Regularization for this Architecture: A neighborhood radius of $\rho=0.05$ may be too large for a relatively small 4-layer character-level LSTM. It may be penalizing the weights too harshly, stripping away useful signal along with the noise.

Slower Convergence: Because SAM computes gradients twice per step, it optimizes more slowly. With early stopping triggered at epoch 10, the model may not have had enough time to fully converge into the new, flatter minimum.

The Nature of Flat Minima: Flat minima inherently sacrifice peak training performance to guarantee stable test performance. In this case, the pendulum swung from overfitting to slight under-optimization.